A certified three-equilibrium bound for 375 fixed quadratic zero-one reaction networks

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Abstract

Consider the fixed list of 375 three-species, six-reaction networks published by Tang and Zhang. For each network in this list and every strictly positive choice of rate constants, we give a computer-assisted proof that there are at most three distinct positive equilibria, including degenerate equilibria. Network-specific elimination certificates yield 272 cubic and 103 quartic residual polynomials. The certificates exclude vanishing elimination denominators at positive equilibria and parameter specializations that would make the residual identically zero; the quartics have opposite signs at their leading and constant coefficients. An independently implemented resultant calculation and a separate Sylvester-determinant check agree for all 375 networks. Within this list, absence of inflow together with acyclicity of the unimolecular production graph selects precisely the two known exceptions to bistability; this is a post-selected finite observation. We also treat one seven-reaction extension, proving at most one exponentially stable positive equilibrium for all positive rates and certifying a rational example with three positive equilibria. The original enumeration and the stability classification of the other 373 networks are outside our independent certification.

1 The finite question and its scope

Tang and Zhang [1] identify 375 quadratic zero-one networks admitting three positive equilibria, of which 373 admit bistability. The classification and the identification of the two exceptions, 33449 and 239089, are theirs. Their Theorem 2 states the upper bound three; Section 5.1 gives an elimination argument for an example, while the Discussion asks for a theoretical explanation of the bound. We address the finite list they supplied, with the rate constants left arbitrary and positive. We are grateful for their classification and data, which make this question accessible.

The contribution is a checked algebraic reduction for every member of that list, covering special positive parameter values as well as generic ones. We do not give a new enumeration of networks or a new general elimination method. Univariate reduction and the relation between the derivative of a reduced equation and the Jacobian determinant belong to established approaches, including Torres–Feliu [2] and Banaji–Feliu [3]. The latter's bounds for specified numbers of distinct reactants do not immediately give three for the present three-species, six-distinct-reactant inputs. Our assertion is a finite collection of all-parameter statements, rather than a test of 375 rate vectors.

Let $Y, N \in \mathbb{Z}^{3\times 6}$ be the reactant and stoichiometric matrices. Column j describes the reaction from $Y_{\cdot j}$ to $Y_{\cdot j} + N_{\cdot j}$ with rate $\kappa_j > 0$. The mass-action equations are

$$\dot{x}_i = F_i(x;\kappa) = \sum_{j=1}^{6} N_{ij}\kappa_j \prod_{\ell=1}^{3} x_\ell^{Y_{\ell j}}. \tag{1}$$

Both complexes have entries zero or one, and every reactant has molecularity at most two. Here "quadratic" means at most two, not that every reaction is bimolecular. All 375 stoichiometric

matrices N have rank three. A positive equilibrium has $x \in \mathbb{R}^3_{>0}$ and $F(x;\kappa) = 0$; it is nondegenerate when $\det J_F \neq 0$, and exponentially stable when all eigenvalues of J_F have negative real part.

Write $\mathcal{N}$ for the 375 records in the fixed upstream file identified in Appendix A. Network numbers below are the upstream `candidate_index`, not row numbers or matrix ranks.

Theorem 1. *For every network $G \in \mathcal{N}$ and every $\kappa \in \mathbb{R}^6_{>0}$, the mass-action system* (1) *has at most three distinct positive equilibria. No nondegeneracy assumption is required.*

Sections 2 and 3 explain the two computational routes supporting this theorem. The structural observation and the seven-reaction proposition are separate results; neither is needed for the upper bound.

2 A univariate certificate for each network

Fix a network and retain one concentration t, writing the other two as u, v. The equations are affine in each of u, v separately, with possible terms $1, u, v, uv$. A recorded invertible integer row transformation $G = TF$ cancels uv in two equations. When uv is absent already, two suitable equations are selected directly. The certificates give

$$A(t)\begin{pmatrix} u \\ v \end{pmatrix} = b(t), \qquad D(t) = \det A(t), \qquad D(t)u = U(t), \quad D(t)v = V(t). \tag{2}$$

All entries are polynomials in t and the rate constants. The record includes T, its nonzero determinant, the coordinate order, D, U, V, and the third transformed equation H.

2.1 Excluding division by zero on the positive branch

In 251 networks, D has nonzero coefficients of one strict sign. In the remaining networks this certificate holds for U (100 networks) or V (24 networks). A nonzero polynomial whose nonzero coefficients have one sign cannot vanish when all its variables are positive.

If a solution of (2) had $D = 0$, the adjugate identities would give $U = V = 0$. Thus a sign certificate for either numerator also excludes $D = 0$ at every positive equilibrium. This step covers special positive rate choices; it is not merely division in a field of rational functions. Every positive equilibrium therefore satisfies

$$u = U/D, \qquad v = V/D,$$

so its retained coordinate t determines it uniquely.

Substitution in H gives the polynomial

$$P(t) = D(t)^2 H\big(U(t)/D(t), V(t)/D(t), t\big) = cL(t)Q(t), \qquad c \in \mathbb{Q} \setminus \{0\}. \tag{3}$$

Every removed factor in L either has coefficients of one strict sign or divides D. Such factors are nonzero at a positive equilibrium. The records contain 706 strict-sign factor entries and 21 divisor-of-D entries, with multiplicities recorded separately. No parameter factor that might vanish at positive rates is discarded without a nonvanishing argument.

Consequently, positive equilibria inject into the positive roots of Q. Conversely, a root of Q with

$$t > 0, \qquad D(t) \neq 0, \qquad U(t)/D(t) > 0, \quad V(t)/D(t) > 0 \tag{4}$$

gives a positive equilibrium: the first two transformed equations hold, $P = 0$ gives $H = 0$, and T is invertible. Roots failing these conditions are not counted as positive equilibria.

Networks	Residual certificate	Consequence at every positive rate vector
272	Degree at most three; at least one coefficient has a strict sign	Q is nonzero and has at most three positive roots.
103	Degree four; leading coefficient strictly negative and constant coefficient strictly positive	At least one negative real root, hence at most three positive roots.

Table 1: Root bounds from the Cramer certificates. Roots are bounded even with multiplicity; the theorem counts distinct equilibria.

2.2 Root bounds that survive specialization

The finite checks give the following two cases.

For a quartic, $Q(0) > 0$ and $Q(t) \to -\infty$ as $t \to -\infty$, so the negative root follows by continuity. For a cubic certificate, a leading coefficient can vanish at particular positive rates, but that only lowers the degree. A recorded strict-sign coefficient prevents the entire polynomial from becoming zero. Thus the argument includes degree drops and degenerate equilibria.

Proof of Theorem 1. For each of the 375 inputs, exact polynomial checks verify the transformation and Cramer identities, the denominator certificate, the factorization (3), the nonvanishing of removed factors at positive equilibria, and the appropriate row of Table 1. Equations (2)–(3) give an injection from positive equilibria to positive roots of a nonzero polynomial with at most three such roots. This proves the assertion for every positive rate vector of each input network. □

The finite calculations here are symbolic. A SymPy construction produces the certificates, and a separately written sparse-polynomial checker using rational arithmetic verifies them without importing SymPy. These two programs were written within one construction effort. The next section describes the independent verification.

3 Independent elimination and certificate checks

An independent implementation starts again from the fixed Y, N matrices. It uses sequential elimination and resultants rather than the two-variable Cramer construction. For an ordered set of coordinates u, v, w, it writes a first equation as

$$F_u = -A(v, w)u + B(v, w), \qquad A(v, w) > 0.$$

It substitutes $u = B/A$ into the remaining equations and clears the positive denominator, obtaining $H(v, w), K(v, w)$. For the selected equation

$$H(v, w) = a(w)v^2 + b(w)v + c(w),$$

the coefficient certificates give $a \le 0$, $c \ge 0$, with at least one inequality strict throughout the positive domain. If both are strict, the two roots have opposite signs. If $c = 0$ there is at most one positive root of the remaining linear factor; if $a = 0$ the nonzero constant term prevents an identically zero linear equation. Hence, for fixed $w > 0$, there is at most one positive v, and then at most one u.

Every positive equilibrium supplies a root of $R(w) = \mathrm{Res}_v(H, K)$. Only factors certified nonzero on the positive domain are removed. Each retained factor has a coefficient whose nonzero monomial coefficients all share one strict sign. This coefficient cannot vanish at positive rates, so the surviving polynomial in w is not identically zero at any positive rate specialization. The records store these tests as `coefficient_signs` within `root_factors`; Appendix A locates

them. Descartes' rule is applied allowing every possible sign, including zero, for each coefficient not assigned a strict sign. The maximum possible number of sign changes bounds all positive roots, including multiple ones. Extra roots introduced by a resultant can only weaken this upper bound; the proved uniqueness of the positive fiber is what permits a bound on equilibria.

This route covers 372 inputs directly. Three require an invertible integer change of equations before the same elimination steps:

Network	First equation	Eliminate; retain	Positive pivot A
48328	$F_1 + F_3$	$x_1, x_2;\ x_3$	$2\kappa_5 x_3$
74814	$F_1 + F_3$	$x_1, x_2;\ x_3$	$\kappa_5 x_3$
100108	$F_1 + F_2$	$x_1, x_3;\ x_2$	$\kappa_6 x_2$

Each transformation has determinant one. The result is again 272 cubic and 103 quartic residuals, all with positive-root bound three. The retained coordinate differs from the Cramer route in 79 inputs, while the degree and conclusion agree in every input.

A second checker in this independent verification reconstructs the substituted equations directly from the original matrices. It expands each Sylvester determinant by permutation enumeration, with sizes at most 4×4, and verifies the factorization and coefficient-sign bounds. It calls neither the first checker's resultant/factorization functions nor its helper functions. The two independent-verification checkers still share SymPy's polynomial arithmetic; this is not a comparison of two unrelated computer-algebra systems.

4 A structural observation about the known exceptions

Call a reaction an *inflow* when its reactant complex is zero. Define the unimolecular production graph on the three species: place $i \to j$, for $i \neq j$, when a reaction whose reactant is exactly X_i produces X_j. This graph uses only unimolecular reactions, not the signs of the full Jacobian.

Observation 2. *Within $\mathcal{N}$, the conjunction of no inflow and an acyclic unimolecular production graph selects exactly networks 33449 and 239089.*

The direct census, independently checked by two cycle-detection procedures, is

	Acyclic	Cyclic	Total
Inflow present	255	112	367
No inflow	2	6	8

The condition was selected after a twelve-network pilot and inspection of the eight no-inflow networks. Its match within this list is therefore not an independent predictive test. Observation 2 is not a theorem that this graph condition determines stability for arbitrary reaction networks. In particular, the graph omits bimolecular reactions.

For the two selected networks, a separate certificate collection [5] gives complete positive-branch reductions: each has at most three distinct positive equilibria and at most one exponentially stable positive equilibrium at arbitrary positive rates. If three distinct positive equilibria exist, exactly one is exponentially stable. We recall these results briefly to identify the known exceptions. The collection is available as a separate data-and-tools release, version 1.0.0 [5].

The determinant criterion of Tang–Zhang, Lemma 3 [1], provides the stability implication for nondegenerate positive equilibria in full-rank three-species zero-one networks. That criterion is prior work. Neither the structural observation nor the bound in Theorem 1 is a new independent certification of the reported bistability classification of all 373 remaining networks.

5 One seven-reaction family

Add the reaction $X_3 \xrightarrow{h} 0$ to network 33449. With the six original rates denoted a, b, c, d, e, f, the equations are

$$\begin{aligned} \dot{x} &= -dx + by + az, \\ \dot{y} &= -by - cyz + exz, \\ \dot{z} &= -(a+h)z + by - cyz + fxy. \end{aligned} \tag{5}$$

All seven rates are strictly positive. This remains a full-rank, three-species quadratic zero-one network.

Proposition 3. *For every $(a, b, c, d, e, f, h) \in \mathbb{R}^7_{>0}$, system (5) has at most three distinct positive equilibria and at most one exponentially stable positive equilibrium.*

Proof. Put $t = z/y$, $L(t) = eat + eb - cd$, and $D(t) = tL(t)$. The first two equilibrium equations give the complete positive branch

$$t > \max\{0, (cd - eb)/(ea)\}, \qquad x = \frac{b(b + at)}{D}, \quad y = \frac{bd}{D}, \quad z = \frac{bd}{L}. \tag{6}$$

Indeed, the first equation gives $x = y(b+at)/d$, and substitution in the second gives $y = bd/(tL)$. Positivity excludes $tL = 0$ and is equivalent to the stated interval. On this branch the third equation is $bdQ_h(t)/D(t)^2$, where

$$\begin{aligned} Q_h(t) = &- ea(a+h)t^3 + [acd - h(eb - cd)]t^2 \\ &+ b(eb + fa - 2cd)t + b^2 f. \end{aligned} \tag{7}$$

Its leading coefficient is strictly negative. Positive equilibria are therefore in one-to-one correspondence with its roots in the interval (6), and there are at most three.

The coordinate change $z = ty$ has determinant y. After this substitution, with t fixed, the (x, y) minor of the first two equations is $-yD$ along their zero set. Differentiating the third equation along the branch and evaluating at $Q_h = 0$ gives

$$\det J = -yQ_h'(t). \tag{8}$$

Exponential stability in three dimensions requires $\det J < 0$, so it requires $Q_h' > 0$. The derivative is a downward-opening quadratic; the set on which it is positive is empty or an interval, and Q_h is strictly increasing there. At most one root can lie in that interval. A root with $Q_h' = 0$ has singular Jacobian and is not exponentially stable. This upper bound uses only a necessary stability condition, not the sufficient direction of Tang–Zhang's lemma. □

The independent derivation retains z instead. With $E(z) = db + (dc - be)z$, the original equations give $ezF_1 + dF_2 = aez^2 - Ey$, so $E > 0$ at every positive equilibrium. They yield

$$x = \frac{az(b + cz)}{E}, \quad y = \frac{aez^2}{E}, \quad P(z) = (a+h)E^2 - aez(b - cz)E - a^2efz^2(b + cz).$$

The remaining equation is $-zP/E^2$ and at an equilibrium $\det J = -zP'/E$. Since $P(0) = (a+h)d^2b^2 > 0$ and $\deg P \le 3$, there can be at most one positive root with $P' > 0$: two upward crossings would require a downward crossing before the first and another between them, hence at least four distinct roots. This gives the same stability bound through a different coordinate.

5.1 An exact rational example

Take

$$(a, b, c, d, e, f, h) = \left(\frac{109}{32768}, \frac{235}{256}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{65536}\right). \tag{9}$$

Here the admissible interval is $t > 0$, and

$$Q_h(t) = -\frac{23871}{4294967296}t^3 + \frac{27797}{33554432}t^2 - \frac{606065}{16777216}t + \frac{55225}{131072}.$$

Exact Sturm sequences isolate one simple root in each interval of Table 2; the total number of admissible roots is three. Rational interval evaluation verifies positive concentrations and the displayed Hurwitz signs, both from the symbolic branch and directly from the original Jacobian. For $\det(\lambda I - J) = \lambda^3 + b_1\lambda^2 + b_2\lambda + b_3$, positivity of $b_1, b_2, b_3, b_1b_2 - b_3$ proves exponential stability. If $b_3 < 0$, the characteristic polynomial has a positive real root.

Root	Rational lower endpoint	Rational upper endpoint	$(b_1, b_2, b_3, b_1b_2 - b_3)$
1	1186617748/63717463	1307453375/70205938	$(+, +, -, +)$
2	1998238606/38640527	3149229703/60897580	$(+, +, +, +)$
3	6820268592/86644919	6713775719/85292030	$(+, +, -, +)$

Table 2: Certified disjoint intervals in increasing t order. Exactly the middle equilibrium is exponentially stable.

The independent z intervals map to these roots in reverse order under $t = db/(aez) + (dc - be)/(ae)$. The exact transformed intervals overlap in intervals containing one root each, and all four Hurwitz signs agree. Thus the comparison checks corresponding roots as well as their counts.

Persistence of nondegenerate equilibria under a sufficiently small dependent reaction addition is consistent with the inheritance results of Banaji–Pantea [4]. The network has full rank, so the added reaction vector is dependent on the original stoichiometric vectors. The family-specific assertion here is the upper bound for *all* positive rates, together with the exact input (9). We make no claim of a new equivalence class or a classification of seven-reaction networks.

6 Verification boundary and limitations

The independent implementation received the fixed upstream data, definitions and statements to test. Its derivations, code and results were fixed before comparison with the construction records. The anticipated conclusions were disclosed, so this was not a blind test of the hypotheses. The independent work used different elimination and a separate implementation; it shared the mathematical definitions, dataset and part of the SymPy arithmetic infrastructure. Roles, dates, hashes and execution records are in the companion `VERIFICATION-RECORD.md`.

The original enumeration yielding the list of 375 and the full existence classification were not independently repeated. Our bound applies exactly to that fixed list and does not by itself prove the same assertion for every three-species, six-reaction quadratic zero-one network. The 373-network bistability classification is outside this independent certification. The structural condition was selected after a pilot, and its extension beyond the list is unresolved. Degenerate nonlinear stability, boundary attractors, periodic orbits, dissipativity and biological realization are not asserted. No human peer review or Lean formalization has been completed.

Use of AI. Codex and Claude assisted derivations, implementation, verification, literature comparison and drafting. Tatsuhiko Takimoto is responsible for the manuscript. The assurances are the mathematical arguments and exact checks specified above.

Claim	Mathematical bridge	Finite verification
375-network bound	Nonvanishing denominator, injective projection, root bound	Cramer identities and signs; independent resultants and Sylvester expansions for every input
Structural observation	Specified graph and finite domain	All 375 graphs checked; precisely two selected
Seven-reaction bound	Complete positive branch and Jacobian/derivative identity	Independent symbolic identity checks
Rational example	Sturm root counting and Hurwitz criterion	Exact isolation, interval signs and root correspondence

Table 3: What supports each assertion.

References

[1] Xiaoxian Tang and Jiandong Zhang. *The Ubiquity of Three Steady States: Minimal Multistable Zero-One Reaction Networks.* arXiv:2608.01116v1, 2026. `https://arxiv.org/abs/2608.01116v1`.

[2] Angélica Torres and Elisenda Feliu. *Symbolic proof of bistability in reaction networks.* arXiv:1909.13608v2, 2020. `https://arxiv.org/abs/1909.13608v2`.

[3] Murad Banaji and Elisenda Feliu. *Positive equilibria in mass action networks: geometry and bounds.* Journal of Mathematical Biology 93, article 37, 2026. Accepted manuscript: arXiv:2409.06877v5. `https://arxiv.org/abs/2409.06877v5`.

[4] Murad Banaji and Casian Pantea. *The inheritance of nondegenerate multistationarity in chemical reaction networks.* arXiv:1608.08400v3, 2017. `https://arxiv.org/abs/1608.08400v3`.

[5] Tatsuhiko Takimoto. *Stability certificates for two exceptional reaction networks.* Data-and-tools collection, version 1.0.0, 2026. Version DOI: `https://doi.org/10.5281/zenodo.23659209`. All-versions DOI: `https://doi.org/10.5281/zenodo.23659208`. Source release: `https://github.com/tkmt-lab/exception-networks-stability-certificates/releases/tag/v1.0.0`.

A Fixed input and certificate index

The upstream input is `results/375_multistable_networks.json` in `https://github.com/zjdong-sudo/paper-multistability-results`, at revision

`323360df12c0787f9bbf7f223a6229b98233be39`.

Its SHA-256 is

`517d1f16f890159185f5384fa2ab089ec4d270e03efe31161ed29489121e252c`.

The companion packet's `manifest.json` records hashes of its files. The 375 candidate identifiers occur in both `certificates/cramer.jsonl` and `certificates/resultants.json`. Paths are relative to the packet root. The latter file combines the independently obtained factors and coefficient-sign records, including the three equation-transform cases. The `certificates/` directory also contains the structural table and rational-example certificates. Run `tools/fetch_input.py` and then `verify.py`; the README gives the environment and commands. The authors' raw data and code are not bundled. The fetch tool obtains the identified input from upstream and checks its SHA-256 before use.

This v1 is a local draft. Before public circulation, the certificate packet and its inventory need a stable versioned public location; no such archive or identifier is asserted here.